Gabriel Freire Ushijima

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Education

University of São Paulo (USP) Jan. 2024 – Dec. 2027

B.S. in Computer Science, IME-USP; GPA: 9.4/10.0 *São Paulo, Brazil*

- Ranked 10th out of 82 students in the entering Computer Science class
- Academic focus: algorithms, computational geometry, mathematical optimization, and programming languages

Experience

Undergraduate Researcher, Touring Polygons Problem Jul. 2025 – Present

IME-USP; FAPESP-funded since Apr. 2026 *São Paulo, Brazil*

- Developed an exact, self-contained solver for the free-order Touring Polygons Problem on non-convex regions
- Compared with Fekete et al. (2026), using their own benchmark instances, my solver was faster on all 558 cases (112× on average) and several orders of magnitude more precise
- Removed dependencies on CGAL and Gurobi, whose licenses restrict commercial use of their solver
- Built an interactive SIICUSP 34 site featuring instances, a challenge, a simulation, and results; presented this research at SIICUSP 33 and 34

Teaching Assistant, Nonlinear Optimization Mar. 2026 – Jul. 2026

Institute of Mathematics, Statistics and Computer Science, University of São Paulo *São Paulo, Brazil*

- Held weekly 2-hour help sessions for a 20-student class, supporting exercises, mathematical formulation, implementation questions, Newton-type methods, line search, and trust regions

Projects

SERIEMA | *Quant AI Itaú Asset 2026, Python*

- Developed and backtested a point-in-time, rainfall-driven long/short strategy for B3 agribusiness stocks, testing whether local weather could anticipate CONAB crop-estimate revisions
- Team ranked among the top 5% in the Desafio Quant AI Itaú Asset 2026

Portal 2D | *Godot*

- Led a four-person USPGameDev team over four months and implemented movement, portals, physics interactions, enemies, UI, and extensible level structure for a playable puzzle game

Morad.IA | *FastAPI, Twilio, LangGraph, Gemini, RAG, ChromaDB*

- Built ChromaDB RAG ingestion/retrieval and Gemini OCR/dossier tools for an AI WhatsApp concierge prototype at a 100-participant Google DeepMind and AI Tinkerers hackathon

Volunteering

Mathematics Tutor Mar. 2026 – Jun. 2026

University Extension Program; Escola Estadual Ibrahim Nobre *São Paulo, Brazil*

- Completed 102 volunteer hours across 34 visits, tutoring high school students in arithmetic, algebra, and problem interpretation

Awards and Competitions

Codewars: Ranked 106th globally and 40th globally in Python among 500k+ active users
Competitions: USP Integration Bee 2024, 2nd place; Maratona USP 2024, 2nd place among freshman teams; OBF 2023 Silver; OBMEP 2023 Bronze; Canguru de Matemática Brasil 2023 Gold

Technical Skills

Languages: Python, C++, C, JavaScript, SQL, Java, Lisp, HTML/CSS, x86-64 Assembly
Frameworks and Tools: FastAPI, Git, CMake, SQLite, WebSockets, WebGL, LangGraph, ChromaDB, Godot, Unreal Engine, CGAL, LaTeX
Mathematics: Algorithms, Computational Geometry, Mathematical Optimization, Nonlinear Optimization, Linear Programming, Branch and Bound
Natural Languages: Portuguese (native), English (advanced)